QUICK CARD · TSM2370 MATERIALTEKNIKK

Lloyd LR 10K – Tensile test

LR 10K Control Software · numbers 1–8 = steps in the software's Quick start

Full guide, videos and software: **scan the QR code** or go to **ravo10.github.io/USN-kongsberg-lab-mech**.

SAFETY

- **Hands out of the grips** while it can move
- **Safety glasses**: specimens can break violently
- Know the **machine's own stop** (F1 needs the PC)
- **Unload (F5)** before you take the specimen out

A. Connect and set up

1–2

1. Machine **on**, serial cable in, start **LR 10K Control Software**.
2. **PC Config.** → COM port → **Connect PC**. The state shows right of the menu bar.
3. **Settings:** tension, crosshead and jog speed, **thickness, width, L0**, units, **load limit** → **Apply to LR 10K**.
4. **File → Specify Storage Folder** (where tests are saved).

B. Clamp and zero

3

5. Move the crosshead with F2 / F3 (hold F4 for fast) until the specimen sits in both grips **without load**.
6. **Set zero point** F6. Without a test the crosshead stops at **50 N**.

Hotkeys

F1	Stop	F6	Set zero point
F2	Up	F7	Start / continue
F3	Down	F8	End test
F4	Fast (hold)	F9	Pause test
F5	Return to zero	Ctrl+S	Save

Troubleshooting

Start Test grey	Its tooltip says what is missing
Will not connect	Check cable and COM port, connect again
"Connection lost"	Plug the adapter back in, Reconnect
E-modulus too low	Normal without extensometer: the grips stretch too

C. Optical extensometer

4a–4d

- Two **green Ø5 mm dots**, **60 mm apart** (placement guide), glued flat on the centre line.
- **View → Extensometer**, camera on. **Pick from image** → click the middle of a dot.
- **L0** = the measured distance between the dots. **Tracking: 2 markers**.
- Even light, no reflections. **Do not touch** the camera or holder during the test.
- If the dots are lost for 0.5 s, the test pauses.

D. Run the test

5

7. **Start Test** F7. Watch the specimen, not only the screen.
8. After fracture: **End Test** F8. The test also ends by itself at the load limit.

E. Results and finish

6–8

9. Check the automatic points: **UTS**, **E-modulus** limits, **Rp0.2**. Ctrl+Click to move or add.
10. **Calculator** → calculate, to see the true stress graph.
11. **Save** Ctrl+S: .lr10k (open again later) + Excel book.
12. **Return to zero point** F5, take the specimen out. **+ New Test** for the next one.

When you leave

- ☐ Broken specimens in the metal recycling box
- ☐ Grips empty, crosshead at a safe height
- ☐ Test saved (and copied to your own storage)
- ☐ Software closed, machine off